

SMART INDIA HACKATHON 2025



Elite Squad

- **Problem Statement ID- 25081**
- **Problem Statement Title- AI-Driven Train Induction Planning & Scheduling for Kochi Metro Rail Limited (KMRL)**
- **Theme- Smart Automation**
- **PS Category- Software**
- **Team ID-86625**
- **Team Name (Registered on portal) - Elite Squad**



AI-Driven Train Induction & Scheduling Platform

Proposed Solution

Current system

- Work is manual and slow (done between 9–11 PM)
- Data is spread across many places (spreadsheets, WhatsApp, logbooks)
- Depends on human guesswork → mistakes happen
- Problems: missed clearances, extra shunting, high cost, delays

Our solution

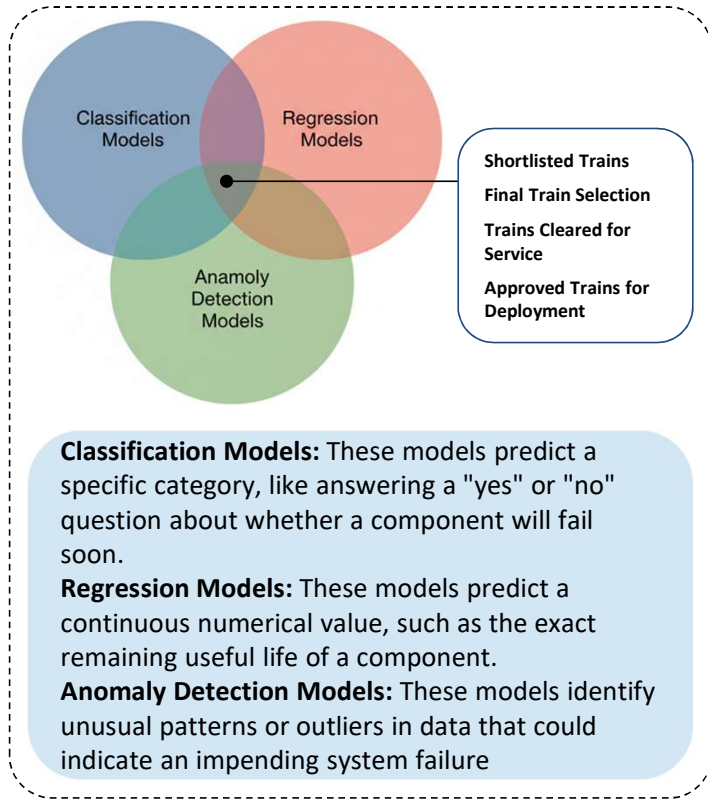
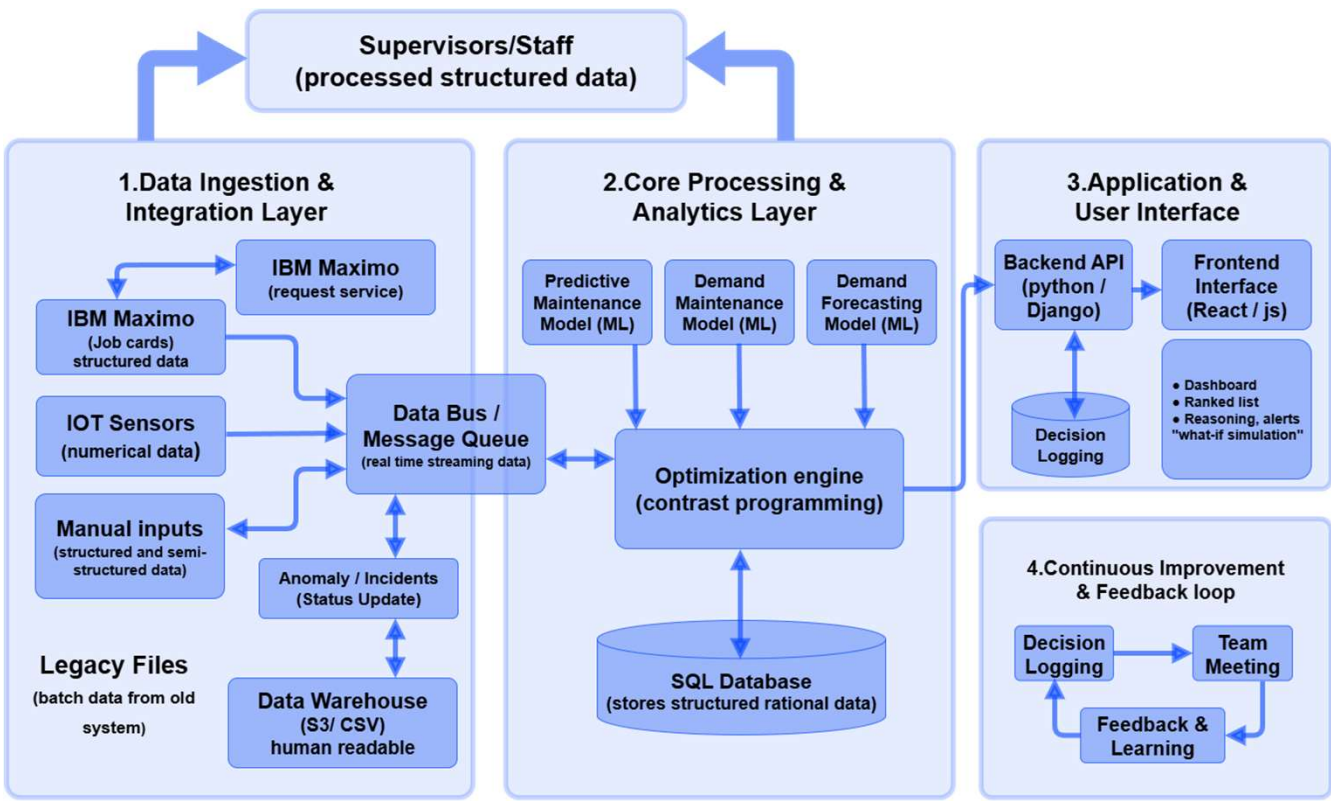
- One platform to collect all data automatically
- Uses simple rules + smart computer models to plan which trains run
- Creates a ready-to-use list of trains for next day
- Shows alerts if something is missing

Why unique

- Brings IoT sensors, Maximo job cards, and manual inputs together
- Learns from past data to get better every day
- Improves safety, punctuality, branding, and cost all at once
- Saves staff time and reduces night-time shunting

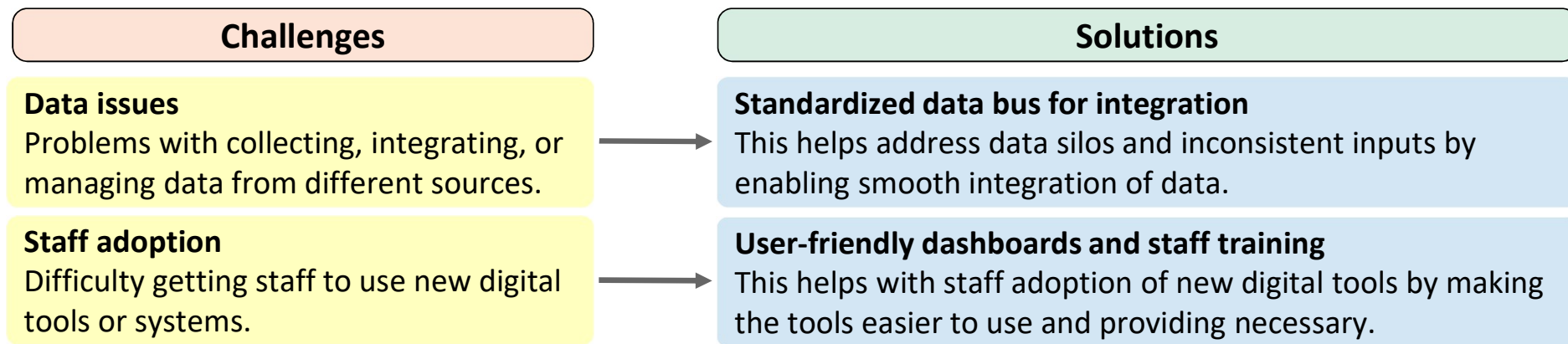


TECHNICAL APPROACH



FEASIBILITY AND VIABILITY

- **Scalable** to handle increasing train operations over time without increasing manpower.
- **Integration Feasibility:** APIs connect existing Maximo & IoT systems.
- **Environmental Feasibility:** Reduced night-time shunting lowers energy consumption and track-occupancy risks, contributing to greener operations.
- **Regulatory Feasibility:** The system makes sure every train follows safety rules, has completed maintenance checks, and meets advertising agreements



IMPACT AND BENEFITS



Impacts

- Improved fleet availability across all train sets
- Maintained 99.5% punctuality through accurate planning
- Reduced energy consumption by minimizing night-time shunting
- Enabled safer and more transparent operations with better data visibility



FLEET AVAILABILITY



99.5%
PUNCTUALITY



REDUCED ENERGY
CONSUMPTION



SAFER & CLEARER
OPERATIONS

Benefits

- Lower maintenance costs through optimized scheduling
- Balanced mileage across trains, extending component life
- Compliance with branding standards, protecting advertiser commitments
- Avoidance of SLA penalties and related revenue loss
- Audit-ready decision-making, ensuring accountability



LOWER
MAINTENANCE COSTS



BALANCED MILEAGE
ACROSS TRAINS



PROTECTION FROM
SLA PENALTIES



COMPLIANCE
WITH BRANDING



AUDIT-READY
ACCOUNTABILITY

RESEARCH AND REFERENCES



- Kochi metro rail limited (KMRL)
<https://kochimetro.org/>
AI-driven predictive maintenance case studies (Indian Railways, BEML).
<https://www.indianrail.gov.in/enquiry/StaticPages/StaticEnquiry.jsp?StaticPage=index.html>
IBM Maximo for Metro Maintenance (official docs).
- **Prototype :**
<https://jayavardhankurasala.github.io/kochimetrops/>

